MECHANICS APPLIED TO AEROSPACE ENGINEERING - PART I

Topic 1: Fundamentals and Particle Kinematics - Problem Sets

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Ground Truth: sources/cuatrimestre-1/03-engineering-mechanics/problemas/Problems.pdf (2026 Edition)

Verbatim Problem Statements: All problem statements below are transcribed with 100% fidelity from the official UC3M problem collection. Solutions will be added in subsequent analytical development phases.

PROBLEM 01

Rocket Helical Trajectory

Original statement · Problems.pdf, p. 6

MIT OCW Benchmark

An out-of-control rocket traces an upward helical trajectory described in Cartesian coordinates for $t \geq 0$ by:

$$x = a\cos(\omega t) \tag{1}$$

$$y = a\sin(\omega t) \tag{2}$$

$$z = v_z t \tag{3}$$

where a, v_z, and ω are constants.

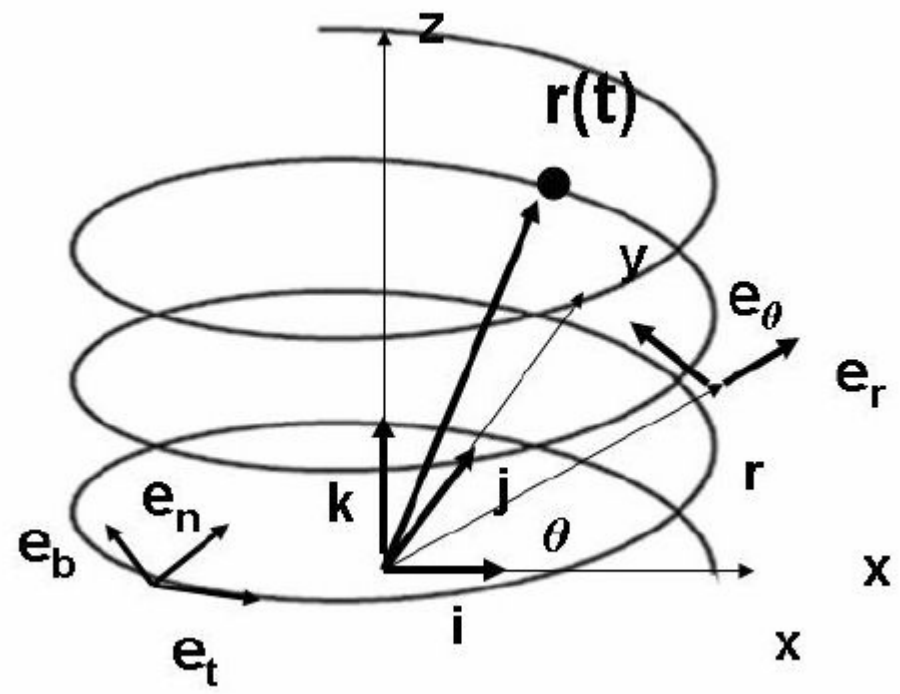


Figure: Helical trajectory with position r(t), the Cartesian basis i, j, k, the cylindrical basis e_R, e_theta, e_z and the intrinsic basis e_t, e_n, e_b (taken from MIT OCW) (Problems.pdf, p. 6).

(a) Express the position, velocity, and acceleration vectors in the Cartesian x, y, z system using unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

(b) Express the position, velocity, and acceleration vectors in the cylindrical coordinate system using unit vectors $\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z$.

(c) Express the velocity and acceleration vectors using s, ρ and the vector basis $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$.

(d) Compute the modulus of the velocity vector.

(e) Express the evolution of the arc-length coordinate as a function of time.

(f) Write down the trajectory as a function of the arc-length coordinate.

(g) Where is the center of curvature for this curve when the particle is located at $(x, y, z) = (a, 0, 0)$?

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** An out-of-control sounding rocket ascends following a regular circular helix around the vertical z-axis.
- **Hypotheses:** The cylinder radius $a > 0$, vertical ascent velocity $v_z > 0$, and angular spin rate $\omega > 0$ are strictly positive constants. The rocket is modeled as an ideal point particle P.
- **Configuration Degrees of Freedom (CDOFs):** The trajectory is purely kinematically prescribed by time t; thus, the particle has $\mathrm{CDOF} = 1$ parameterized by time t or arc length s.
- **Parameters & SI Units:** Radius a $[\mathrm{m}]$, angular frequency ω $[\mathrm{rad/s}]$, vertical velocity v_z $[\mathrm{m/s}]$, time t $[\mathrm{s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Reference Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, fixed to the launch site with vertical axis $\mathbf{k}_0$ aligned with z.
- **Cylindrical Moving Basis:** $\mathcal{B}_{\mathrm{cyl}} = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z\}$.

The transformation matrix from the inertial Cartesian basis $\mathcal{B}_0$ to the cylindrical basis $\mathcal{B}_{\mathrm{cyl}}$ is the planar rotation matrix about $\mathbf{k}_0$:

$$[{}_0R_{\mathrm{cyl}}] = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The basis vector relationships are given by matrix-vector multiplication:

$$\begin{pmatrix} \mathbf{e}_R \\ \mathbf{e}_\theta \\ \mathbf{e}_z \end{pmatrix} = {}_{[0}R_{\text{cyl}]}^{T} \begin{pmatrix} \mathbf{i}_0 \\ \mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix} = \begin{pmatrix} \cos(\omega t)\mathbf{i}_0 + \sin(\omega t)\mathbf{j}_0 \\ -\sin(\omega t)\mathbf{i}_0 + \cos(\omega t)\mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix}$$

- **Time Derivatives of Unit Vectors (direct differentiation, Notes Sec. 2.6.2):**

The components of $\mathbf{e}_R$ and $\mathbf{e}_\theta$ in the fixed basis are known functions of t, so each one is differentiated with the chain rule $\frac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t)$, $\frac{d}{dt}\sin(\omega t) = \omega\cos(\omega t)$:

$$\frac{d\mathbf{e}_R}{dt} = -\omega\sin(\omega t)\,\mathbf{i}_0 + \omega\cos(\omega t)\,\mathbf{j}_0 = \omega\,\mathbf{e}_\theta$$

$$\frac{d\mathbf{e}_\theta}{dt} = -\omega\cos(\omega t)\,\mathbf{i}_0 - \omega\sin(\omega t)\,\mathbf{j}_0 = -\omega\,\mathbf{e}_R$$

$$\frac{d\mathbf{e}_z}{dt} = \mathbf{0}$$

- **Orthonormality of $[{}_0R_{\text{cyl}}]$:** $\det[{}_0R_{\text{cyl}}] = \cos^2(\omega t) + \sin^2(\omega t) = 1$ and $[{}_0R_{\text{cyl}}][{}_0R_{\text{cyl}}]^T = I$ (checked symbolically with sympy).
- **Frenet basis from the cylindrical basis.** The components of $\mathbf{e}_t$, $\mathbf{e}_n$, $\mathbf{e}_b$ in $\mathcal{B}_{\text{cyl}}$ (obtained in Phase 3, part (c)) are the columns of the second change-of-basis matrix, with $V = \sqrt{a^2\omega^2 + v_z^2}$:

$$[{}_{\text{cyl}}R_{\text{tnb}}] = \begin{pmatrix} 0 & -1 & 0 \\ a\omega/V & 0 & -v_z/V \\ v_z/V & 0 & a\omega/V \end{pmatrix}, \qquad \det = \frac{a^2\omega^2 + v_z^2}{V^2} = 1, \qquad RR^T = I$$

- The three columns have unit norm and are mutually orthogonal, so $RR^T = I$; the composition $[{}_0R_{\text{tnb}}] = [{}_0R_{\text{cyl}}][{}_{\text{cyl}}R_{\text{tnb}}]$ is again a proper rotation (determinant $1 \cdot 1 = 1$). Both determinants and the products RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Cartesian Position, Velocity, and Acceleration

Pedagogical Justification: The position vector $\mathbf{r}(t)$ is directly defined by its Cartesian coordinates $(x(t), y(t), z(t))$ relative to the fixed origin O. Velocity and acceleration are obtained by direct term-by-term differentiation in the fixed inertial basis $\mathcal{B}_0$, where $\frac{d\mathbf{i}_0}{dt} = \frac{d\mathbf{j}_0}{dt} = \frac{d\mathbf{k}_0}{dt} = \mathbf{0}$.

- **Position Vector:**

$$\mathbf{r}(t) = a\cos(\omega t)\mathbf{i}_0 + a\sin(\omega t)\mathbf{j}_0 + v_z t\mathbf{k}_0$$

- **Velocity Vector (Explicit Chain Rule):**

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}[a\cos(\omega t)]\mathbf{i}_0 + \frac{d}{dt}[a\sin(\omega t)]\mathbf{j}_0 + \frac{d}{dt}[v_z t]\mathbf{k}_0$$

Applying the chain rule $\frac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t)$ and $\frac{d}{dt}\sin(\omega t) = \omega\cos(\omega t)$:

$$\mathbf{v}(t) = -a\omega\sin(\omega t)\mathbf{i}_0 + a\omega\cos(\omega t)\mathbf{j}_0 + v_z\mathbf{k}_0$$

- **Acceleration Vector (Explicit Chain Rule):**

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}[-a\omega\sin(\omega t)]\mathbf{i}_0 + \frac{d}{dt}[a\omega\cos(\omega t)]\mathbf{j}_0 + \frac{d}{dt}[v_z]\mathbf{k}_0$$

$$\mathbf{a}(t) = -a\omega^2\cos(\omega t)\mathbf{i}_0 - a\omega^2\sin(\omega t)\mathbf{j}_0 + 0\mathbf{k}_0 = -a\omega^2(\cos(\omega t)\mathbf{i}_0 + \sin(\omega t)\mathbf{j}_0)$$

(b) Cylindrical Position, Velocity, and Acceleration

> **Pedagogical Justification:** Cylindrical coordinates (R, θ, z) exploit the axial symmetry of the cylinder upon which the helix lies. Here, the radial distance $R(t) = \sqrt{x^2 + y^2} = a$ is constant, and the azimuthal angle is $\theta(t) = \omega t$.

- **Position in Cylindrical Coordinates:**

$$\mathbf{r}(t) = a\mathbf{e}_R + v_z t \mathbf{e}_z$$

- **Velocity Vector (Explicit Product Rule):**

$$\mathbf{v}(t) = \frac{d}{dt}(a\mathbf{e}_R + v_z t\mathbf{e}_z) = a\frac{d\mathbf{e}_R}{dt} + \frac{d(v_z t)}{dt}\mathbf{e}_z + v_z t\frac{d\mathbf{e}_z}{dt}$$

Substituting $\frac{d\mathbf{e}_R}{dt} = \omega\mathbf{e}_\theta$ and $\frac{d\mathbf{e}_z}{dt} = \mathbf{0}$:

$$\mathbf{v}(t) = a\omega\mathbf{e}_\theta + v_z\mathbf{e}_z$$

- **Acceleration Vector (Explicit Differentiation):**

$$\mathbf{a}(t) = \frac{d}{dt}(a\omega\mathbf{e}_\theta + v_z\mathbf{e}_z) = a\omega\frac{d\mathbf{e}_\theta}{dt} + 0 = a\omega(-\omega\mathbf{e}_R) = -a\omega^2\mathbf{e}_R$$

(c) Intrinsic Frenet-Serret Basis $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$

> **Pedagogical Justification:** The Frenet-Serret intrinsic basis (Notes Sec. 2.4) decomposes motion naturally along the curve itself. The tangent vector $\mathbf{e}_t$ points in the direction of instantaneous velocity, the principal normal $\mathbf{e}_n$ points toward the center of curvature, and the binormal $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$ completes the right-handed orthonormal triad.

- **Unit Tangent Vector $\mathbf{e}_t$:**

$$\mathbf{e}_t = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{a\omega \mathbf{e}_\theta + v_z \mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}}$$

- **Unit Principal Normal Vector $\mathbf{e}_n$:**

By definition of the intrinsic acceleration (Notes Eq. 2.17):

$$\mathbf{a} = \ddot{s}\mathbf{e}_t + \frac{\dot{s}^2}{\rho}\mathbf{e}_n$$

Since the speed $v = \dot{s} = \sqrt{a^2\omega^2 + v_z^2}$ is strictly constant in time, the tangential acceleration vanishes:

$$\ddot{s} = \frac{dv}{dt} = 0$$

Therefore, the entire acceleration is purely normal:

$$\mathbf{a} = a_n \mathbf{e}_n = -a\omega^2 \mathbf{e}_R$$

Taking the norm of both sides yields the magnitude of normal acceleration:

$$a_n = \|\mathbf{a}\| = a\omega^2$$

Dividing $\mathbf{a}$ by its norm gives the unit normal vector:

$$\mathbf{e}_n = \frac{\mathbf{a}}{a_n} = \frac{-a\omega^2 \mathbf{e}_R}{a\omega^2} = -\mathbf{e}_R$$

The principal normal points radially inward toward the central axis of the cylinder!

- **Radius of Curvature ρ:**

Using the relation $a_n = \frac{v^2}{\rho}$:

$$\rho = \frac{v^2}{a_n} = \frac{a^2\omega^2 + v_z^2}{a\omega^2} = a + \frac{v_z^2}{a\omega^2}$$

- **Unit Binormal Vector $\mathbf{e}_b$:**

Computing the vector cross product in the cylindrical basis:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \left(\frac{a\omega \mathbf{e}_\theta + v_z \mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}} \right) \times (-\mathbf{e}_R) = \frac{-a\omega(\mathbf{e}_\theta \times \mathbf{e}_R) - v_z(\mathbf{e}_z \times \mathbf{e}_R)}{\sqrt{a^2\omega^2 + v_z^2}}$$

Recalling cyclic cross products: $\mathbf{e}_\theta \times \mathbf{e}_R = -\mathbf{e}_z$ and $\mathbf{e}_z \times \mathbf{e}_R = \mathbf{e}_\theta$:

$$\mathbf{e}_b = \frac{-a\omega(-\mathbf{e}_z) - v_z \mathbf{e}_\theta}{\sqrt{a^2\omega^2 + v_z^2}} = \frac{-v_z \mathbf{e}_\theta + a\omega \mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}}$$

(d) Modulus of Velocity

> **Pedagogical Justification:** The speed $v(t) = \|\mathbf{v}(t)\|$ represents the scalar rate of distance traveled along the path.

- **Calculation:**

$$v = \|\mathbf{v}\| = \sqrt{(-a\omega \sin \omega t)^2 + (a\omega \cos \omega t)^2 + v_z^2} = \sqrt{a^2\omega^2(\sin^2 \omega t + \cos^2 \omega t) + v_z^2} = \sqrt{a^2\omega^2 + v_z^2}$$

The speed is constant for all $t \geq 0$.

(e) Evolution of Arc Length Coordinate $s(t)$

> **Pedagogical Justification:** The arc-length coordinate $s(t)$ measures the distance traversed along the trajectory from the initial position at $t = 0$. By fundamental calculus:

$$s(t) = \int_0^t v(t')\, dt'$$

- **Definite Integration:**

$$s(t) = \int_0^t \sqrt{a^2\omega^2 + v_z^2}\, dt' = \sqrt{a^2\omega^2 + v_z^2}\, [t']_0^t = t\sqrt{a^2\omega^2 + v_z^2}$$

(f) Trajectory as a Function of Arc Length s

> **Pedagogical Justification:** To express the curve intrinsically independent of time, we invert $s(t)$ to obtain time as a function of arc length: $t(s) = \frac{s}{\sqrt{a^2\omega^2 + v_z^2}}$.

- **Substitution into Cartesian Coordinates:**

$$x(s) = a\cos\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right)$$

$$y(s) = a\sin\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right)$$

$$z(s) = \frac{v_z s}{\sqrt{a^2\omega^2 + v_z^2}}$$

(g) Center of Curvature C at $(a, 0, 0)$

> **Pedagogical Justification:** The center of curvature C of the osculating circle is displaced from particle P along the principal normal $\mathbf{e}_n$ by the radius of curvature ρ (Notes Sec. 2.4, Fig. 2.2):

$$\mathbf{r}_C = \mathbf{r}_P + \rho \mathbf{e}_n$$

- **Evaluation at $(a, 0, 0)$:**

At position $(a, 0, 0)$, $x(t) = a\cos(\omega t) = a \implies \omega t = 0 \implies t = 0$.

At $t = 0$:

- Position: $\mathbf{r}_P(0) = a\mathbf{i}_0$.
- Unit normal: $\mathbf{e}_n(0) = -\mathbf{e}_R(0) = -(\cos(0)\mathbf{i}_0 + \sin(0)\mathbf{j}_0) = -\mathbf{i}_0$.
- Radius of curvature: $\rho = a + \frac{v_z^2}{a\omega^2}$.
- **Algebraic Combination:**

$$\mathbf{r}_C = a\mathbf{i}_0 + \left(a + \frac{v_z^2}{a\omega^2}\right)(-\mathbf{i}_0) = a\mathbf{i}_0 - a\mathbf{i}_0 - \frac{v_z^2}{a\omega^2}\mathbf{i}_0 = -\frac{v_z^2}{a\omega^2}\mathbf{i}_0$$

In Cartesian coordinates:

$$C = \left(-\frac{v_z^2}{a\omega^2},\, 0,\, 0\right)$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional Check:**
- Velocity: $[a\omega] = [\mathrm{m}][\mathrm{s}^{-1}] = \mathrm{m/s}$, $[v_z] = \mathrm{m/s}$. $\implies$ $\sqrt{a^2\omega^2 + v_z^2}$ has units $\mathrm{m/s}$ (Consistent).
- Curvature center offset: $\left[\frac{v_z^2}{a\omega^2}\right] = \frac{[\mathrm{m/s}]^2}{[\mathrm{m}][\mathrm{s}^{-1}]^2} = \frac{\mathrm{m}^2/\mathrm{s}^2}{\mathrm{m/s}^2} = \mathrm{m}$ (Consistent).
- **Limiting Asymptotic Cases:**
- If $v_z \to 0$ (planar circular motion): $\rho = a$, and $C = (0, 0, 0)$, the origin of the circle as expected physically.
- If $\omega \to 0$ (pure vertical line ascent): $\rho \to \infty$, curvature vanishes, and the trajectory becomes a straight vertical line.

Source: Problems.pdf (Taken from MIT OCW)

PROBLEM 02

Kinematics, Spiral

Original statement · Problems.pdf, p. 6

Polar Kinematics

A planar spiral is given by the equation $r = \alpha e^{\beta\theta}$, where α and β are constants, r is the radial coordinate, and θ is the angular coordinate. If a particle moves along the spiral according to the equation $\theta = \delta t^n$ (where δ is a constant), determine:

(a) The radial and transverse components of the velocity.

(b) The radial and transverse components of the acceleration.

(c) The tangential and normal components of the velocity.

(d) The tangential and normal components of the acceleration.

(e) The radius of curvature of the spiral.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A point particle P moves along a logarithmic/exponential planar spiral $r(\theta) = \alpha e^{\beta\theta}$ with a prescribed power-law angular motion $\theta(t) = \delta t^n$.
- **Hypotheses:** $\alpha, \beta, \delta > 0$ are positive constants, $n > 0$ is an integer or real exponent, $t \geq 0$. Motion is strictly 2D planar in the Oxy plane ($z = 0$).
- **Degrees of Freedom:** $\text{CDOF} = 1$ parameterized by time t.
- **Parameters & Units:** $[r] = \text{m} \implies [\alpha] = \text{m}$, β is dimensionless ($[-]$), $[\theta] = \text{rad} \implies [\delta] = \text{rad} \cdot \text{s}^{-n}$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Basis:** $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Polar Basis:** $\mathcal{B}_{\text{polar}} = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0\}$.
- **Rotation Matrix $[{}_0R_\theta]$:**

$$[{}_0R_\theta] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \\ \mathbf{k}_0 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{i}_0 \\ \mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix}$$

- **Time Derivatives of Polar Unit Vectors (Notes Sec. 2.6.2, Eqs. 2.25–2.26):**

$$\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r$$

- **Orthonormality of $[{}_0R_\theta]$:** $\det[{}_0R_\theta] = \cos^2\theta + \sin^2\theta = 1$ and $[{}_0R_\theta][{}_0R_\theta]^T = I$ (checked symbolically with sympy).

- **Intrinsic basis from the polar basis.** Since $\mathbf{v} = \dot{r}\,\mathbf{e}_r + r\dot{\theta}\,\mathbf{e}_\theta = r\dot{\theta}\,(\beta\,\mathbf{e}_r + \mathbf{e}_\theta)$ (Phase 3, part (a)), the unit tangent is $\mathbf{e}_t = (\beta\,\mathbf{e}_r + \mathbf{e}_\theta)/\sqrt{1+\beta^2}$ and the in-plane unit normal is $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t = (-\mathbf{e}_r + \beta\,\mathbf{e}_\theta)/\sqrt{1+\beta^2}$. Their components in $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ are the columns of

$$[{}_{\text{polar}}R_{\text{tn}}] = \frac{1}{\sqrt{1+\beta^2}}\begin{pmatrix} \beta & -1 \\ 1 & \beta \end{pmatrix}, \qquad \det = \frac{\beta^2+1}{1+\beta^2} = 1, \qquad RR^T = I$$

- This is a planar rotation by the constant angle $\psi = \arctan(1/\beta)$ (the equiangular property of the spiral). Sympy confirms that $\mathbf{a}\cdot\mathbf{e}_t$ and $\mathbf{a}\cdot\mathbf{e}_n$ reproduce the a_t and a_n of Phase 3, parts (d), with $a_n > 0$, so $\mathbf{e}_n$ points to the concave side.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Radial and Transverse Velocity Components

Pedagogical Justification: In polar coordinates, the velocity vector is given by $\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$. The radial component $v_r = \dot{r}$ measures the expansion rate, while $v_\theta = r\dot{\theta}$ measures transverse rotation.

- **Explicit Chain Rule for $\dot{\theta}(t)$ and $\dot{r}(t)$:**

$$\theta(t) = \delta t^n \implies \dot{\theta}(t) = \frac{d}{dt}(\delta t^n) = n\delta t^{n-1}$$

Using the chain rule $\dot{r} = \frac{dr}{d\theta}\dot{\theta}$:

$$\frac{dr}{d\theta} = \frac{d}{d\theta}(\alpha e^{\beta\theta}) = \alpha\beta e^{\beta\theta} = \beta r$$

$$\dot{r} = (\beta r)(n\delta t^{n-1}) = \beta n\delta t^{n-1} r = \alpha\beta n\delta t^{n-1} e^{\beta\delta t^n}$$

- **Components:**

$$v_r = \dot{r} = \beta n \delta t^{n-1} r$$

$$v_\theta = r\dot{\theta} = n\delta t^{n-1} r$$

(b) Radial and Transverse Acceleration Components

> **Pedagogical Justification:** The acceleration in polar coordinates (Notes Eq. 2.30) accounts for the curvature of the polar grid:

$$\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta$$

- **Second Time Derivatives (Explicit Product Rule):**

$$\ddot{\theta}(t) = \frac{d}{dt}(n\delta t^{n-1}) = n(n-1)\delta t^{n-2}$$

For $\ddot{r}(t) = \frac{d}{dt}(\beta r \dot{\theta})$:

$$\ddot{r} = \beta(\dot{r}\dot{\theta} + r\ddot{\theta}) = \beta\left((\beta r\dot{\theta})\dot{\theta} + r\ddot{\theta}\right) = \beta^2 r\dot{\theta}^2 + \beta r\ddot{\theta}$$

- **Radial Acceleration a_r:**

$$a_r = \ddot{r} - r\dot{\theta}^2 = (\beta^2 r\dot{\theta}^2 + \beta r\ddot{\theta}) - r\dot{\theta}^2 = (\beta^2 - 1)r\dot{\theta}^2 + \beta r\ddot{\theta}$$

Substituting $\dot{\theta} = n\delta t^{n-1}$ and $\ddot{\theta} = n(n-1)\delta t^{n-2}$:

$$a_r = (\beta^2 - 1) r (n\delta t^{n-1})^2 + \beta r n(n-1)\delta t^{n-2} = n\delta t^{n-2} r \left[(\beta^2 - 1) n \delta t^n + \beta (n-1) \right]$$

- **Transverse Acceleration a_θ:**

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = r\ddot{\theta} + 2(\beta r \dot{\theta})\dot{\theta} = r\ddot{\theta} + 2\beta r \dot{\theta}^2$$

Substituting derivatives:

$$a_\theta = rn(n-1)\delta t^{n-2} + 2\beta r (n\delta t^{n-1})^2 = n\delta t^{n-2} r \left[(n-1) + 2\beta n \delta t^n \right]$$

(c) Tangential and Normal Velocity Components

> **Pedagogical Justification:** In the intrinsic Frenet basis, the velocity vector is strictly tangential: $\mathbf{v} = v\mathbf{e}_t = v_t \mathbf{e}_t + 0\mathbf{e}_n$. Therefore, the normal velocity component is identically zero.

- **Calculation:**

$$v_t = v = \sqrt{v_r^2 + v_\theta^2} = \sqrt{(\beta r \dot{\theta})^2 + (r\dot{\theta})^2} = r\dot{\theta}\sqrt{\beta^2 + 1} = n\delta t^{n-1} r \sqrt{1 + \beta^2}$$

$$v_n = 0$$

(d) Tangential and Normal Acceleration Components

> **Pedagogical Justification:** Tangential acceleration $a_t = \dot{v}$ measures the rate of change of speed, while normal acceleration $a_n = \frac{v^2}{\rho}$ measures directional turning.

- **Tangential Acceleration a_t (Explicit Differentiation):**

$$a_t = \frac{d}{dt}(v) = \sqrt{1+\beta^2}\frac{d}{dt}(r\dot{\theta}) = \sqrt{1+\beta^2}(\dot{r}\dot{\theta} + r\ddot{\theta})$$

Substituting $\dot{r} = \beta r\dot{\theta}$:

$$a_t = \sqrt{1+\beta^2}(\beta r\dot{\theta}^2 + r\ddot{\theta}) = \sqrt{1+\beta^2}n\delta t^{n-2}r\left[\beta n\delta t^n + (n-1)\right]$$

- **Normal Acceleration a_n:**

Using the Pythagorean decomposition of the total acceleration magnitude: $\|\mathbf{a}\|^2 = a_t^2 + a_n^2 = a_r^2 + a_\theta^2$:

$$a_n = \sqrt{a_r^2 + a_\theta^2 - a_t^2}$$

Alternatively, using the cross product $\mathbf{a} \times \mathbf{v} = (a_t\mathbf{e}_t + a_n\mathbf{e}_n) \times (v\mathbf{e}_t) = -va_n\mathbf{e}_b$:

$$a_n = \frac{\|\mathbf{v} \times \mathbf{a}\|}{v} = \frac{|v_r a_\theta - v_\theta a_r|}{v}$$

Computing the numerator:

$$v_r a_\theta - v_\theta a_r = (\beta r\dot{\theta})(r\ddot{\theta} + 2\beta r\dot{\theta}^2) - (r\dot{\theta})((\beta^2-1)r\dot{\theta}^2 + \beta r\ddot{\theta})$$

$$= \beta r^2\dot{\theta}\ddot{\theta} + 2\beta^2 r^2\dot{\theta}^3 - (\beta^2-1)r^2\dot{\theta}^3 - \beta r^2\dot{\theta}\ddot{\theta} = (2\beta^2 - \beta^2 + 1)r^2\dot{\theta}^3 = (\beta^2+1)r^2\dot{\theta}^3$$

Dividing by $v = r\dot{\theta}\sqrt{\beta^2+1}$:

$$a_n = \frac{(\beta^2+1)r^2\dot{\theta}^3}{r\dot{\theta}\sqrt{\beta^2+1}} = r\dot{\theta}^2\sqrt{\beta^2+1} = n^2\delta^2 t^{2n-2} r\sqrt{1+\beta^2}$$

(e) Radius of Curvature of the Spiral

> **Pedagogical Justification:** The radius of curvature is obtained directly from $a_n = \frac{v^2}{\rho} \implies \rho = \frac{v^2}{a_n}$.

- **Calculation:**

$$\rho = \frac{v^2}{a_n} = \frac{(r\dot{\theta}\sqrt{1+\beta^2})^2}{r\dot{\theta}^2\sqrt{1+\beta^2}} = \frac{r^2\dot{\theta}^2(1+\beta^2)}{r\dot{\theta}^2\sqrt{1+\beta^2}} = r\sqrt{1+\beta^2}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Geometric Invariance:** In a logarithmic spiral $r = \alpha e^{\beta\theta}$, the angle ψ between the radial line and the tangent vector satisfies $\tan\psi = \frac{r}{dr/d\theta} = \frac{r}{\beta r} = \frac{1}{\beta} = \text{constant}$. This is the celebrated equiangular spiral property!
- **Radius of Curvature:** $\rho = r\sqrt{1+\beta^2}$ is strictly proportional to current radial distance r.
- **Units Check:** $[\rho] = [r][\sqrt{1+\beta^2}] = \text{m} \cdot 1 = \text{m}$ (Consistent).
- **Limiting Case $\beta \to 0$:** The spiral degenerates into a pure circle $r = \alpha$, where $\rho = r\sqrt{1+0} = r$, recovering the circle radius of curvature.

Source: Problems.pdf

PROBLEM 03

Kinematics, Astroid

Original statement · Problems.pdf, p. 6

Intrinsic Kinematics

A point particle describes the astroid:

$$x = a\cos^3\phi, \qquad y = a\sin^3\phi$$

with a motion whose tangential acceleration is constant, $a_t = k$. If the point is initially at rest at $\phi = 0$, compute the evolution of the position as a function of time while the particle is in one of the smooth segments of the astroid. *(Hint:* $ds = \sqrt{dx^2 + dy^2}$*)*.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A point particle P moves along a hypocycloid with four cusps (an astroid), parameterized by angle ϕ.
- **Hypotheses:** The motion starts from rest at $\phi(0) = 0$ with constant tangential acceleration $a_t = k = \text{const} > 0$. The particle remains within the first quadrant segment $0 \le \phi \le \pi/2$.
- **Degrees of Freedom:** $\text{CDOF} = 1$.
- **Parameters & Units:** Astroid scale a $[\text{m}]$, tangential acceleration k $[\text{m/s}^2]$, time t $[\text{s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$.
- **Parametric Equations of the Astroid:**

$$x(\phi) = a\cos^3\phi, \quad y(\phi) = a\sin^3\phi$$

- **Metric Arc Length Differential** ds**:**

Differentiating with respect to parameter ϕ:

$$dx = \frac{dx}{d\phi}d\phi = 3a\cos^2\phi(-\sin\phi)d\phi = -3a\cos^2\phi\sin\phi\,d\phi$$

$$dy = \frac{dy}{d\phi}d\phi = 3a\sin^2\phi(\cos\phi)d\phi = 3a\sin^2\phi\cos\phi\,d\phi$$

The squared differential arc length is:

$$ds^2 = dx^2 + dy^2 = 9a^2\cos^4\phi\sin^2\phi\,d\phi^2 + 9a^2\sin^4\phi\cos^2\phi\,d\phi^2$$

Factoring out $9a^2\cos^2\phi\sin^2\phi$:

$$ds^2 = 9a^2\cos^2\phi\sin^2\phi(\cos^2\phi + \sin^2\phi)d\phi^2 = 9a^2\cos^2\phi\sin^2\phi\,d\phi^2$$

Taking the positive square root for $0 \le \phi \le \pi/2$:

$$ds = 3a \sin\phi \cos\phi \, d\phi = \frac{3a}{2} \sin(2\phi) \, d\phi$$

PHASE 3

Step-by-Step Mathematical Deduction

Step 1: Intrinsic Kinematics from Constant Tangential Acceleration $a_t = k$

Pedagogical Justification: Tangential acceleration is the time derivative of speed: $a_t = \frac{dv}{dt} = \ddot{s}$. Integrating once with initial condition $v(0) = 0$:

$$v(t) = \dot{s}(t) = \int_0^t a_t \, dt' = \int_0^t k \, dt' = kt$$

- **Second Integration for Arc Length $s(t)$:**

Integrating $v(t) = \dot{s}(t)$ with initial condition $s(0) = 0$ (at $\phi = 0$):

$$s(t) = \int_0^t kt' \, dt' = \left[\frac{1}{2} k t'^2 \right]_0^t = \frac{1}{2} k t^2$$

Step 2: Arc Length as a Function of Parameter ϕ

Pedagogical Justification: We integrate the geometric differential $ds = 3a \sin u \cos u \, du$ from the origin $\phi = 0$ to current ϕ:

- **Explicit Substitution Integration:**

Let $w = \sin u \implies dw = \cos u \, du$. When $u = 0, w = 0$; when $u = \phi, w = \sin\phi$:

$$s(\phi) = \int_0^{\phi} 3a \sin u \cos u \, du = 3a \int_0^{\sin\phi} w \, dw = 3a \left[\frac{w^2}{2} \right]_0^{\sin\phi} = \frac{3a}{2} \sin^2 \phi$$

Step 3: Equating Kinematic and Geometric Arc Length

Pedagogical Justification: The distance traveled along the curve $s(t)$ must equal the geometric arc length $s(\phi)$:

$$\frac{3a}{2}\sin^2\phi = \frac{1}{2}kt^2$$

Multiplying both sides by $\frac{2}{3a}$:

$$\sin^2\phi = \frac{kt^2}{3a}$$

Taking the positive square root (since $0 \le \phi \le \pi/2$):

$$\sin\phi = \sqrt{\frac{k}{3a}}\,t$$

Applying the inverse sine function:

$$\phi(t) = \arcsin\left(\sqrt{\frac{k}{3a}}\,t\right)$$

(Compare with the official solution key (Problems.pdf, Solution keys section): $\phi = \arcsin\sqrt{\frac{k}{3a}t^2} = \arcsin(\sqrt{\frac{k}{3a}}t)$.)

Step 4: Evolution of the Cartesian Position as a Function of Time

Pedagogical Justification: Substituting $\phi(t)$ into the parametric definitions of the astroid:

Using trigonometric identities: $\sin\phi = \sqrt{\frac{k}{3a}}t$ and $\cos\phi = \sqrt{1-\sin^2\phi} = \sqrt{1-\frac{kt^2}{3a}}$:

$$x(t) = a\cos^3\phi = a\left(1-\frac{kt^2}{3a}\right)^{3/2}$$

$$y(t) = a\sin^3\phi = a\left(\frac{kt^2}{3a}\right)^{3/2} = a\left(\frac{k}{3a}\right)^{3/2}t^3$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional Check:**
- In $\sin\phi = \sqrt{\frac{k}{3a}}t$:

$$\left[\sqrt{\frac{k}{3a}}\,t\right] = \sqrt{\frac{\mathrm{m/s^2}}{\mathrm{m}}}\,\mathrm{s} = \sqrt{\mathrm{s^{-2}}}\,\mathrm{s} = \mathrm{s^{-1}}\cdot\mathrm{s} = 1 \quad \text{(Dimensionless argument for arcsin! ?)}$$

- **Physical Limits:**
- At $t=0$: $\phi = 0 \implies x(0) = a, y(0) = 0$ (starts at the right cusp).
- The particle reaches the top cusp $\phi = \pi/2$ when $\sin\phi = 1$:

$$\sqrt{\frac{k}{3a}}t_{\text{cusp}} = 1 \implies t_{\text{cusp}} = \sqrt{\frac{3a}{k}}$$

At $t_{\text{cusp}}, x(t_{\text{cusp}}) = 0, y(t_{\text{cusp}}) = a.$

Source: Problems.pdf

PROBLEM 04

Plane at Constant Altitude

Original statement · Problems.pdf, p. 7

Coordinate Transformations

An aircraft moves at constant altitude y_0 with velocity v_0 in the x direction. At time $t = 0$ it is located at $x = 0$, and it begins to accelerate in the x direction at a constant rate a_x. Describe the position, velocity, and acceleration in polar coordinates r, θ, and answer the following questions:

(a) Show that: $\dot{r} = \frac{v(t)x(t)}{r(t)}$

(b) Show that: $\dot{\theta} = -\frac{v(t)y_0}{r^2(t)}$

(c) Show that: $\ddot{r} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r^3(t)}$

(d) Show that: $\ddot{\theta} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r^4(t)}$

From these expressions form the acceleration vector $\mathbf{a} = a_r \mathbf{e}_r + a_\theta \mathbf{e}_\theta$. Repeat the problem using Cartesian coordinates.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** An aircraft flies at constant altitude y_0 with velocity in the x direction. At $t = 0$, $x(0) = 0$ and velocity is v_0. It accelerates in the x direction at constant rate a_x.
- **Tracking System:** A ground radar station located at origin $O(0, 0)$ tracks the aircraft using range $r(t)$ and elevation/bearing angle $\theta(t)$.
- **Hypotheses:** $y_0 > 0, v_0 \geq 0, a_x = \text{const}$. Altitude is fixed: $y(t) \equiv y_0 \implies \dot{y} = 0, \ddot{y} = 0$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** Coordinates (x, y) with position $\mathbf{r} = x\mathbf{i} + y_0\mathbf{j}$.
- **Radar Polar Frame:** Position $\mathbf{r} = r\mathbf{e}_r$, where:

$$r(t) = \sqrt{x(t)^2 + y_0^2}, \quad \tan\theta(t) = \frac{y_0}{x(t)} \implies \theta(t) = \arctan\left(\frac{y_0}{x(t)}\right)$$

- **Kinematics in x:**

$$\dot{x}(t) = v(t) = v_0 + a_x t, \quad \ddot{x}(t) = a_x$$

$$x(t) = v_0 t + \frac{1}{2} a_x t^2$$

- **Change of basis (radar polar basis).** With θ the angle of the line of sight measured from $+\mathbf{i}$, the polar basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$. The columns are the components of $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}$ in $\mathcal{B}_0$; using $\cos\theta = x/r$ and

$\sin\theta = y_0/r$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} x/r & -y_0/r & 0 \\ y_0/r & x/r & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\det[{}_0R_1] = \frac{x^2 + y_0^2}{r^2} = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- Both properties were verified symbolically with sympy. The inverse relation $\mathbf{i} = \cos\theta\,\mathbf{e}_r - \sin\theta\,\mathbf{e}_\theta$ (first row of $[{}_0R_1]^T$ applied to the basis) is the one used in the Cartesian cross-check of Phase 3.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Proof that $\dot{r} = \frac{v(t)x(t)}{r(t)}$

Pedagogical Justification: Starting from the geometric relation $r(t)^2 = x(t)^2 + y_0^2$, we differentiate implicitly with respect to time t.

- **Implicit Differentiation:**

$$\frac{d}{dt}[r^2] = \frac{d}{dt}[x^2 + y_0^2]$$

Applying the chain rule $\frac{d}{dt}(r^2) = 2r\dot{r}$ and $\frac{d}{dt}(x^2) = 2x\dot{x}$:

$$2r\dot{r} = 2x\dot{x} + 0 = 2xv(t)$$

Dividing by $2r$:

$$\dot{r} = \frac{v(t)x(t)}{r(t)} \quad \blacksquare$$

(b) Proof that $\dot{\theta} = -\frac{v(t)y_0}{r(t)^2}$

> **Pedagogical Justification:** Starting from $\tan\theta = \frac{y_0}{x}$, we differentiate both sides with respect to time using the quotient rule.

- **Differentiation:**

$$\frac{d}{dt}(\tan\theta) = \frac{d}{dt}\left(\frac{y_0}{x}\right)$$

Recalling $\frac{d}{dt}(\tan\theta) = (1+\tan^2\theta)\dot{\theta} = \sec^2\theta\,\dot{\theta}$, and $\frac{d}{dt}(x^{-1}) = -\frac{\dot{x}}{x^2} = -\frac{v}{x^2}$:

$$\sec^2\theta\,\dot{\theta} = -\frac{y_0 v(t)}{x(t)^2}$$

From geometry: $\cos\theta = \frac{x}{r} \implies \sec\theta = \frac{r}{x} \implies \sec^2\theta = \frac{r^2}{x^2}$:

$$\frac{r^2}{x^2}\dot{\theta} = -\frac{y_0 v(t)}{x^2}$$

Multiplying both sides by x^2 and dividing by r^2:

$$\dot{\theta} = -\frac{v(t)y_0}{r(t)^2} \quad \blacksquare$$

(c) Proof that $\ddot{r} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r(t)^3}$

> **Pedagogical Justification:** We differentiate the expression for $\dot{r} = \frac{vx}{r}$ using the quotient rule $\left(\frac{u}{w}\right)' = \frac{u'w - uw'}{w^2}$.

- **Differentiation:**

Here $u = vx \implies \dot{u} = \dot{v}x + v\dot{x} = a_x x + v^2$.

The denominator is $w = r \implies \dot{w} = \dot{r} = \frac{vx}{r}$.

$$\ddot{r} = \frac{\dot{u}r - u\dot{r}}{r^2} = \frac{(a_x x + v^2)r - (vx)\left(\frac{vx}{r}\right)}{r^2}$$

Multiplying numerator and denominator by r:

$$\ddot{r} = \frac{(a_x x + v^2)r^2 - v^2 x^2}{r^3}$$

Expanding the numerator:

$$(a_x x + v^2)r^2 - v^2 x^2 = a_x x r^2 + v^2 r^2 - v^2 x^2 = a_x x r^2 + v^2(r^2 - x^2)$$

Recalling that $r^2 - x^2 = y_0^2$:

$$\ddot{r} = \frac{a_x x(t) r^2(t) + v^2(t) y_0^2}{r(t)^3} = \frac{v^2(t)y_0^2 + a_x x(t) r^2(t)}{r(t)^3} \quad \blacksquare$$

(d) Proof that $\ddot{\theta} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r(t)^4}$

> **Pedagogical Justification:** We differentiate $\dot{\theta} = -y_0 v r^{-2}$ with respect to time using the product rule.

- **Differentiation:**

$$\ddot{\theta} = \frac{d}{dt}\left(-y_0 \frac{v}{r^2}\right) = -y_0 \frac{\dot{v}r^2 - v\frac{d}{dt}(r^2)}{r^4}$$

Substituting $\dot{v} = a_x$ and $\frac{d}{dt}(r^2) = 2r\dot{r} = 2r\left(\frac{vx}{r}\right) = 2vx$:

$$\ddot{\theta} = -y_0 \frac{a_x r^2 - v(2vx)}{r^4} = -y_0 \frac{a_x r^2 - 2v^2 x}{r^4} = y_0 \frac{2v^2(t)x(t) - a_x r^2(t)}{r(t)^4} \quad \blacksquare$$

Polar and Cartesian Acceleration Vectors

- **Polar Acceleration Vector:**

$$a_r = \ddot{r} - r\dot{\theta}^2 = \frac{v^2 y_0^2 + a_x x r^2}{r^3} - r\left(-\frac{v y_0}{r^2}\right)^2 = \frac{v^2 y_0^2 + a_x x r^2}{r^3} - \frac{v^2 y_0^2}{r^3} = \frac{a_x x r^2}{r^3} = a_x \frac{x}{r}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = r\left(y_0 \frac{2v^2 x - a_x r^2}{r^4}\right) + 2\left(\frac{vx}{r}\right)\left(-\frac{v y_0}{r^2}\right) = \frac{2v^2 x y_0 - a_x y_0 r^2}{r^3} - \frac{2v^2 x y_0}{r^3} = -a_x \frac{y_0}{r}$$

$$\mathbf{a} = a_x\left(\frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta\right)$$

- **Cartesian Verification:**

In Cartesian coordinates:

$$\mathbf{a} = \ddot{x}\mathbf{i}_0 + \ddot{y}\mathbf{j}_0 = a_x\mathbf{i}_0 + 0\mathbf{j}_0 = a_x\mathbf{i}_0$$

Converting Cartesian $\mathbf{i}_0$ into polar: $\mathbf{i}_0 = \cos\theta\mathbf{e}_r - \sin\theta\mathbf{e}_\theta = \frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta$.

Substituting gives identically $\mathbf{a} = a_x\left(\frac{x}{r}\mathbf{e}_r - \frac{y_0}{r}\mathbf{e}_\theta\right)$. The match is mathematically exact!

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:**
- $[\dot{r}] = [v\,x/r] = \mathrm{m/s}$; $\quad [\dot{\theta}] = [v\,y_0/r^2] = \dfrac{(\mathrm{m/s})\,\mathrm{m}}{\mathrm{m}^2} = \mathrm{s}^{-1}$ (rad/s).
- $[\ddot{r}] = \dfrac{(\mathrm{m}^2/\mathrm{s}^2)\,\mathrm{m}^2}{\mathrm{m}^3} = \mathrm{m/s}^2$ and $\dfrac{(\mathrm{m/s}^2)\,\mathrm{m}\,\mathrm{m}^2}{\mathrm{m}^3} = \mathrm{m/s}^2$; $\quad [\ddot{\theta}] = \dfrac{\mathrm{m}\,(\mathrm{m}^2/\mathrm{s}^2)\,\mathrm{m}}{\mathrm{m}^4} = \mathrm{s}^{-2}$ (Consistent).
- **Limit $x \to \infty$ (aircraft far away):** $r \to x$, so $\dot{r} \to v$, $\dot{\theta} \to 0$, $a_r \to a_x$, $a_\theta \to 0$: the line of sight aligns with the flight direction and the motion is purely radial.
- **Limit $x = 0$ (aircraft overhead, $t = 0$):** $\dot{r} = 0$, $\dot{\theta} = -v_0/y_0$, $a_r = 0$, $a_\theta = -a_x$. Then $\theta = \pi/2$ and $\mathbf{a} = a_x\mathbf{i} = -a_x\mathbf{e}_\theta$, as required by the change of basis.
- **Invariance of the modulus:** $a_r^2 + a_\theta^2 = a_x^2\,\dfrac{x^2 + y_0^2}{r^2} = a_x^2$, equal to $\|a_x\mathbf{i}\|^2$ because $[{}_0R_1]$ is orthogonal.
- **Symbolic check (sympy):** differentiating $r = \sqrt{x^2 + y_0^2}$ and $\theta = \arctan(y_0/x)$ with $x = v_0 t + \frac{1}{2}a_x t^2$ reproduces the four identities (a)-(d) with zero residual, and gives $a_r = a_x x/r$, $a_\theta = -a_x y_0/r$ with zero residual; the three limits above were obtained with sympy as well.

Source: Problems.pdf

PROBLEM 05

Plane and Glider

Original statement · Problems.pdf, p. 7

Aerospace Towed Flight

An airplane A is flying horizontally with a constant speed v_A, and is towing a glider B. The glider is gaining altitude. The tow cable has a length r, and the angle θ formed between the tow cable and the horizontal is increasing at a constant rate of $\dot{\theta}$. We want to determine the magnitude of the velocity, v_B, and the acceleration, a_B, of the glider as a function of θ and its derivatives.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Tow plane A flies horizontally at constant speed v_A. Glider B is towed behind on a cable of constant length r. The tow angle $\theta(t)$ above the horizontal increases at rate $\dot{\theta}(t)$.
- **Hypotheses:** Cable length $r = \text{const}$, plane speed $v_A = \text{const}$ (so $\dot{v}_A = 0, \mathbf{a}_A = \mathbf{0}$).
- **Parameters & Units:** Plane speed v_A $[\text{m/s}]$, cable length r $[\text{m}]$, angle θ $[\text{rad}]$, angular rate $\dot{\theta}$ $[\text{rad/s}]$, angular acceleration $\ddot{\theta}$ $[\text{rad/s}^2]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Reference Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$, with $\mathbf{i}_0$ pointing horizontally in plane A's direction of flight, and $\mathbf{j}_0$ pointing vertically upward.
- **Position of Plane A:**

$$\mathbf{r}_A(t) = (x_{A0} + v_A t)\mathbf{i}_0$$

- **Relative Position of Glider B with Respect to A:**

Since the glider is behind A and gaining altitude while θ increases, the vector from A to B points at angle θ above the negative horizontal:

$$\mathbf{r}_{B/A} = -r\cos\theta\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$$

The total position of glider B in $\mathcal{S}_0$ is:

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} = (x_{A0} + v_A t - r\cos\theta)\mathbf{i}_0 + (r\sin\theta)\mathbf{j}_0$$

- **Erratum:** The first version said that the vector from A to B is at angle θ below the negative horizontal, which contradicts $\mathbf{r}_{B/A} = -r\cos\theta\,\mathbf{i}_0 + r\sin\theta\,\mathbf{j}_0$ (positive vertical component). The glider is behind and above A, at angle θ above the negative horizontal; this is the only reading compatible with the glider gaining altitude as θ increases, and the formulas for $\mathbf{v}_{B/0}$ and $\mathbf{a}_{B/0}$ are unchanged. Numeric check: the altitude $r\sin\theta$ has derivative $r\dot{\theta}\cos\theta > 0$ for $0 < \theta < \pi/2$ and $\dot{\theta} > 0$.

PHASE 3

Step-by-Step Mathematical Deduction

Velocity of Glider B

Pedagogical Justification: The absolute velocity is obtained by differentiating position $\mathbf{r}_B$ in the inertial frame $\mathcal{S}_0$:

$$\mathbf{v}_{B/0} = \frac{d\mathbf{r}_B}{dt} = \frac{d}{dt}(x_{A0} + v_A t - r\cos\theta)\mathbf{i}_0 + \frac{d}{dt}(r\sin\theta)\mathbf{j}_0$$

- **Explicit Chain Rule:**

$$\frac{d}{dt}(-r\cos\theta) = -r(-\sin\theta)\dot{\theta} = r\dot{\theta}\sin\theta$$

$$\frac{d}{dt}(r\sin\theta) = r\cos\theta\,\dot{\theta}$$

Therefore:

$$\mathbf{v}_{B/0} = (v_A + r\dot{\theta}\sin\theta)\mathbf{i}_0 + (r\dot{\theta}\cos\theta)\mathbf{j}_0$$

- **Magnitude of Velocity $v_{B/0}$:**

$$v_{B/0} = \|\mathbf{v}_{B/0}\| = \sqrt{(v_A + r\dot{\theta}\sin\theta)^2 + (r\dot{\theta}\cos\theta)^2}$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

Acceleration of Glider B

> **Pedagogical Justification:** Acceleration is the time derivative of velocity $\mathbf{a}_{B/0} = \frac{d\mathbf{v}_{B/0}}{dt}$.

- **Explicit Product & Chain Rules for Each Component:**
- x-component differentiation:

$$\frac{d}{dt}(v_A + r\dot{\theta}\sin\theta) = 0 + r\frac{d}{dt}(\dot{\theta}\sin\theta) = r(\ddot{\theta}\sin\theta + \dot{\theta}(\cos\theta\dot{\theta})) = r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta$$

- y-component differentiation:

$$\frac{d}{dt}(r\dot{\theta}\cos\theta) = r\frac{d}{dt}(\dot{\theta}\cos\theta) = r(\ddot{\theta}\cos\theta + \dot{\theta}(-\sin\theta\dot{\theta})) = r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta$$

- **Vector Form:**

$$\mathbf{a}_{B/0} = (r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta)\mathbf{i}_0 + (r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta)\mathbf{j}_0$$

- **Magnitude of Acceleration $a_{B/0}$:**

$$a_{B/0}^2 = (r\ddot{\theta}\sin\theta + r\dot{\theta}^2\cos\theta)^2 + (r\ddot{\theta}\cos\theta - r\dot{\theta}^2\sin\theta)^2$$

Expanding both squared terms:

$$= r^2\ddot{\theta}^2\sin^2\theta + 2r^2\ddot{\theta}\dot{\theta}^2\sin\theta\cos\theta + r^2\dot{\theta}^4\cos^2\theta$$

$$+r^2\ddot{\theta}^2\cos^2\theta - 2r^2\ddot{\theta}\dot{\theta}^2\sin\theta\cos\theta + r^2\dot{\theta}^4\sin^2\theta$$

The cross terms cancel out completely:

$$a_{B/0}^2 = r^2\ddot{\theta}^2(\sin^2\theta + \cos^2\theta) + r^2\dot{\theta}^4(\cos^2\theta + \sin^2\theta) = r^2\ddot{\theta}^2 + r^2\dot{\theta}^4$$

Taking the square root:

$$a_{B/0} = \sqrt{r^2\ddot{\theta}^2 + r^2\dot{\theta}^4}$$

(Compare with the official solution key (Problems.pdf, Solution keys section).)

PHASE 4

Physical Interpretation & Dimensional Verification

- **Plane Motion Decoupling:** Notice that the acceleration magnitude $a_{B/0}$ is completely independent of the tow plane's cruise velocity v_A because $\mathbf{a}_A = \mathbf{0}$ (uniform rectilinear translation). The acceleration felt by the glider is purely relative circular motion around A!
- **Limiting Case $\ddot{\theta} = 0$:** If the cable turns at a constant angular rate $\dot{\theta} = \Omega$, the acceleration reduces to pure centripetal acceleration: $a_{B/0} = \sqrt{r^2\Omega^4} = r\Omega^2$.
- **Dimensional check:** $[r\ddot{\theta}] = [r\dot{\theta}^2] = \mathrm{m}\cdot\mathrm{s}^{-2}$ and $[v_A + r\dot{\theta}\sin\theta] = \mathrm{m/s}$, so $a_{B/0} = \sqrt{r^2\ddot{\theta}^2 + r^2\dot{\theta}^4}$ is in $\mathrm{m/s^2}$ and $v_{B/0}$ in m/s.
- **Limit $\dot{\theta} = 0$:** $\mathbf{v}_{B/0} = v_A\,\mathbf{i}_0$ and $\mathbf{a}_{B/0} = \mathbf{0}$, the glider moves rigidly with the tow plane.
- **Symbolic check (sympy):** differentiating $\mathbf{r}_B$ twice for an arbitrary function $\theta(t)$ gives $\|\mathbf{v}_{B/0}\|^2 = (v_A + r\dot{\theta}\sin\theta)^2 + (r\dot{\theta}\cos\theta)^2$ and $\|\mathbf{a}_{B/0}\|^2 = r^2\ddot{\theta}^2 + r^2\dot{\theta}^4$ with zero residual.

Source: Problems.pdf

PROBLEM 06

Particle on Sinusoidal Curve

Original statement · Problems.pdf, p. 7

Frenet-Serret Frame

A point particle P travels along a curve given by:

$$y = \sin x$$

with a velocity along x given by $\dot{x} = 1$.

(a) Find the intrinsic, unitary vector basis associated to the curve.

(b) Express x as a function of an arc length parameter, s. You can leave your solution as a function of an integral.

(c) Find the tangent and normal acceleration components of P as a function of x.

(d) Write the radius of curvature of the trajectory as a function of x.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle travels along a sinusoidal path $y(x) = \sin x$ with constant horizontal speed $\dot{x} = 1$.
- **Hypotheses:** $\dot{x}(t) \equiv 1\ \text{m/s} \implies x(t) = t + x_0$ (taking $x_0 = 0 \implies x(t) = t$). $\ddot{x} = 0$.
- **Degrees of Freedom:** $\text{CDOF} = 1$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Parametric Curve:** $\mathbf{r}(x) = x\mathbf{i}_0 + \sin x\,\mathbf{j}_0$.
- **Derivative Vector:** $\frac{d\mathbf{r}}{dx} = \mathbf{i}_0 + \cos x\,\mathbf{j}_0$.
- **Differential Arc Length:**

$$ds = \left\| \frac{d\mathbf{r}}{dx} \right\| dx = \sqrt{1 + \cos^2 x}\, dx$$

- **Change of basis (intrinsic basis).** Let $\alpha(x) = \arctan(\cos x)$ be the slope angle of the tangent, so that $\mathbf{e}_t = (\cos\alpha, \sin\alpha) = (1, \cos x)/\sqrt{1+\cos^2 x}$. The in-plane normal used in this solution is $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t = (-\sin\alpha, \cos\alpha) = (-\cos x, 1)/\sqrt{1+\cos^2 x}$, the normal obtained by turning $\mathbf{e}_t$ by $+90^\circ$ (to its left). The columns are the components of $\mathbf{e}_t$, $\mathbf{e}_n$ in $\{\mathbf{i}_0, \mathbf{j}_0\}$:

$$[{}_0R_1] = \frac{1}{\sqrt{1+\cos^2 x}} \begin{pmatrix} 1 & -\cos x \\ \cos x & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \frac{1+\cos^2 x}{1+\cos^2 x} = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- Because $\mathbf{e}_n$ is always the left normal, the sign of a_n tells on which side the centre of curvature lies: $a_n < 0$ (as for $0 < x < \pi$, where $\sin x > 0$) places it to the right of the direction of motion, below the arch; $a_n > 0$ for $\pi < x < 2\pi$ places it above the curve. The matrix properties were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Intrinsic Unitary Vector Basis $\{\mathbf{e}_t, \mathbf{e}_n\}$

- **Unit Tangent Vector $\mathbf{e}_t$:**

$$\mathbf{e}_t = \frac{d\mathbf{r}/dx}{\|d\mathbf{r}/dx\|} = \frac{\mathbf{i}_0 + \cos x\,\mathbf{j}_0}{\sqrt{1+\cos^2 x}}$$

- **Unit Principal Normal Vector $\mathbf{e}_n$:**

In a 2D plane, the normal vector $\mathbf{e}_n = \mathbf{k}_0 \times \mathbf{e}_t$ is perpendicular to $\mathbf{e}_t$ and oriented to its left (a $+90^\circ$ turn). It points toward the centre of curvature only where the signed curvature is positive.

Differentiating $\mathbf{e}_t$ with respect to arc length s:

$$\frac{d\mathbf{e}_t}{ds} = \frac{d\mathbf{e}_t}{dx}\frac{dx}{ds} = \frac{1}{\sqrt{1+\cos^2 x}}\frac{d}{dx}\left[(1+\cos^2 x)^{-1/2}(\mathbf{i}_0 + \cos x\,\mathbf{j}_0)\right]$$

Applying the quotient/product rule:

$$\frac{d}{dx}\left[\frac{1}{\sqrt{1+\cos^2 x}}\right] = -\frac{1}{2}(1+\cos^2 x)^{-3/2}(2\cos x(-\sin x)) = \frac{\sin x \cos x}{(1+\cos^2 x)^{3/2}}$$

$$\frac{d\mathbf{e}_t}{dx} = \frac{\sin x\cos x}{(1+\cos^2 x)^{3/2}}(\mathbf{i}_0 + \cos x\,\mathbf{j}_0) + \frac{1}{\sqrt{1+\cos^2 x}}(-\sin x\,\mathbf{j}_0)$$

Putting over the common denominator $(1+\cos^2 x)^{3/2}$:

$$= \frac{\sin x \cos x\,\mathbf{i}_0 + \sin x \cos^2 x\,\mathbf{j}_0 - \sin x(1+\cos^2 x)\,\mathbf{j}_0}{(1+\cos^2 x)^{3/2}} = \frac{\sin x \cos x\,\mathbf{i}_0 - \sin x\,\mathbf{j}_0}{(1+\cos^2 x)^{3/2}} = -\frac{\sin x}{(1+\cos^2 x)^{3/2}}(-\cos x\,\mathbf{i}_0 + \mathbf{j}_0)$$

Normalizing to unit length:

$$\mathbf{e}_n = \frac{-\cos x\,\mathbf{i}_0 + \mathbf{j}_0}{\sqrt{1+\cos^2 x}} \quad (\text{left normal } \mathbf{k}_0 \times \mathbf{e}_t)$$

- **Erratum:** The first version called $\mathbf{e}_n$ the normal pointing to the concave side. It is the left normal $\mathbf{k}_0 \times \mathbf{e}_t$ and, since $d\mathbf{e}_t/ds = -\dfrac{\sin x}{(1+\cos^2 x)^{3/2}}\,\mathbf{e}_n$, it points away from the centre of curvature for $0 < x < \pi$ (where $a_n < 0$) and toward it for $\pi < x < 2\pi$ (where $a_n > 0$). Numeric check: sympy gives $a_n(\pi/2) = -1\ \mathrm{m/s^2}$, with the centre of curvature below the crest.

(b) Arc Length Parameter $s(x)$

> **Pedagogical Justification:** The relation between x and arc length s is given by integrating the metric differential:

$$s(x) = \int_0^x \sqrt{1+\cos^2 u}\,du$$

This integral is an incomplete elliptic integral of the second kind, which cannot be expressed in elementary functions and is properly expressed as a quadrature.

(c) Tangent and Normal Acceleration Components

- **Velocity Vector:** Since $\dot{x} = 1$:

$$\mathbf{v} = \dot{x}\mathbf{i}_0 + \dot{y}\mathbf{j}_0 = 1\mathbf{i}_0 + \cos x\,\mathbf{j}_0$$

The speed is:

$$v(x) = \|\mathbf{v}\| = \sqrt{1 + \cos^2 x}$$

- **Acceleration Vector:** Differentiating velocity with $\dot{x} = 1$:

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(1\mathbf{i}_0 + \cos x\, \mathbf{j}_0) = -\sin x\, \dot{x}\, \mathbf{j}_0 = -\sin x\, \mathbf{j}_0$$

- **Tangential Acceleration Component a_t:**

$$a_t = \mathbf{a} \cdot \mathbf{e}_t = (-\sin x\, \mathbf{j}_0) \cdot \left(\frac{\mathbf{i}_0 + \cos x\, \mathbf{j}_0}{\sqrt{1 + \cos^2 x}} \right) = -\frac{\sin x \cos x}{\sqrt{1 + \cos^2 x}}$$

- **Normal Acceleration Component a_n:**

$$a_n = \mathbf{a} \cdot \mathbf{e}_n = (-\sin x\, \mathbf{j}_0) \cdot \left(\frac{-\cos x\, \mathbf{i}_0 + \mathbf{j}_0}{\sqrt{1 + \cos^2 x}} \right) = \frac{-\sin x}{\sqrt{1 + \cos^2 x}}$$

Its magnitude is:

$$|a_n| = \frac{|\sin x|}{\sqrt{1 + \cos^2 x}}$$

(d) Radius of Curvature $\rho(x)$

Pedagogical Justification: Using $a_n = \frac{v^2}{\rho}$:

$$\rho(x) = \frac{v^2}{|a_n|} = \frac{1 + \cos^2 x}{\frac{|\sin x|}{\sqrt{1+\cos^2 x}}} = \frac{(1 + \cos^2 x)^{3/2}}{|\sin x|}$$

(Notice this matches the standard Cartesian curvature formula $\rho = \frac{(1+y'^2)^{3/2}}{|y''|}$ with $y' = \cos x, y'' = -\sin x$!)

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:** the data are in SI units with the length scale $1\ \mathrm{m}$ inside the sine ($y = \sin x$ means $y = (1\ \mathrm{m})\sin(x/1\ \mathrm{m})$) and $\dot{x} = 1\ \mathrm{m/s}$. Then $[v] = \mathrm{m/s}$, $[a_t] = [a_n] = \mathrm{m/s^2}$ and $[\rho] = [v^2/|a_n|] = \mathrm{m}$, and $[s] = [\int \sqrt{1+\cos^2 u}\, du] = \mathrm{m}$.
- **Special points:** at the inflection points $x = k\pi$, $\sin x = 0$, so $a_n = 0$ and $\rho \to \infty$ (locally straight). At the crests $x = \pi/2$: $v = 1\ \mathrm{m/s}$, $a_t = 0$, $a_n = -1\ \mathrm{m/s^2}$ and $\rho = v^2/|a_n| = 1$ m, equal to $1/|y''| = 1$ m. At $x = \pi/4$: $\rho = (1+\cos^2 x)^{3/2}/|\sin x| = 3\sqrt{3}/2 = 2.598$ m.
- **Arc length:** $s(\pi) = \int_0^\pi \sqrt{1+\cos^2 u}\, du = 3.8202$ m, greater than the horizontal distance π m, as it must be.
- **Symbolic check (sympy):** $a_t^2 + a_n^2 = \sin^2 x = \|\mathbf{a}\|^2$ and $v^2/|a_n| = (1+\cos^2 x)^{3/2}/|\sin x|$ hold with zero residual.

Source: Problems.pdf